

Roll No.

TEE-601

**B. TECH. (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

POWER ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Latching and Holding currents as applicable to an SCR. Show these current on its static V-I characteristics. (CO1, CO2)
- (b) Define the following terms related to the SCR and discuss their significance : (CO1, CO2)
 - (i) Forward Break over voltage
 - (ii) Peak Inverse Voltage
 - (iii) On-state voltage Drop
 - (iv) Critical Rate of rise of Voltage
 - (v) Delay time for turn on the SCR

P. T. O.

- (c) Explain the significance of snubber circuit. Explain the importance of considering resistance in combination with capacitor. (CO1, CO2)
2. (a) Explain Single phase full wave SCR circuit feeds power to DC motor load. Draw the wave form of source voltage, load voltage, load current, and voltage across the SCR for the firing angle 60 degree. (CO1, CO2)
- (b) Describe the working of three phase semi converter. Discuss whether the freewheeling diode comes into play or not. Also, obtain an expression for the average output voltage. (CO1, CO2)
- (c) A single phase one pulse converter with RLE load has the following data :
- Supply voltage = 230 V at 50 Hz, $R = 2 \text{ Ohm}$, $L = 1 \text{ mH}$, $E = 120 \text{ V}$.
Extinction angle is 220 degree and firing angle is 25 degree. (CO1, CO2)
- (i) Calculate the voltage across the SCR at the instant SCR is triggered.
- (ii) Find the voltage that appears across SCR when current decays to zero.
- (iii) Find the Peak inverse voltage for the SCR.
3. (a) Describe the working of Single Phase full bridge Inverter. Explain how the output power of full bridge inverter becomes four times the power handled by single phase half bridge inverter. (CO3)
- (b) Explain the principle of working of three phase bridge inverter with appropriate circuit diagram. Draw phase and line voltage for 120 degree mode of conduction and R-load is connected as star connected. (CO3)

(3)

- (c) In multiple pulse width modulation used in the inverter, the amplitude and the frequency for the triangular carrier and square wave are respectively 4V, 6 kHz and 1V, 1 kHz. Calculate the number of pulses per half cycle and pulse width. Also calculate the RMS output voltage when in supply by 200 V. (CO3)
4. (a) Describe the principle of step-up chopper. Derive an expression for the average output voltage in terms of input voltage and duty cycle. (CO4)
- (b) Describe the working of type D chopper with appropriate waveform to demonstrate its operation in first as well as fourth quadrant. (CO4)
- (c) What is DC chopper. Describe the working of type B chopper. Does it operate as a step down or step up chopper? (CO4)
5. (a) What is cycloconverter? Explain single phase to single phase step-up cycloconverter. Make the conduction of various thyristor. (CO5, CO6)
- (b) Compare the possible configuration of single phase voltage controller. (CO5, CO6)
- (c) A 230 V, 1 kW electric heater is fed through a TRIAC from 230 V, 50 Hz source. Find the load power for firing angle delay of 70 degree. Derive the expression used for the voltage. (CO5, CO6)

Roll No.

TEE-602

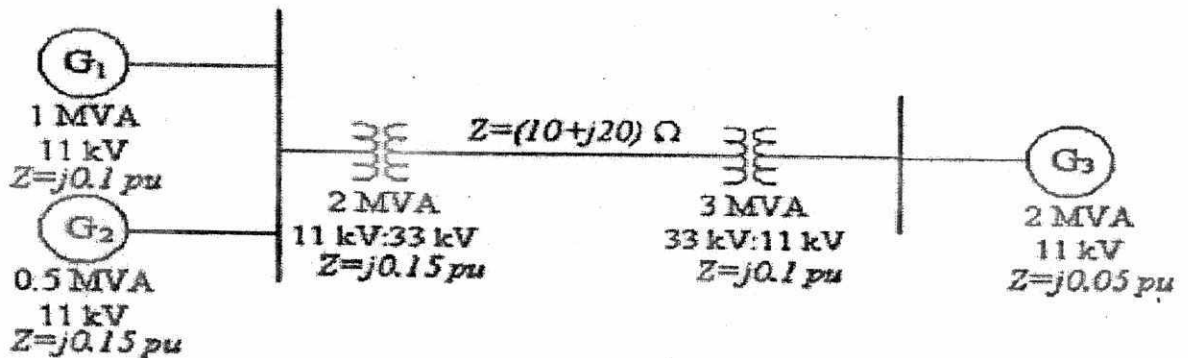
**B. TECH. (EE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

POWER SYSTEM-II

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Define the terms per voltage, per unit impedance and per unit volt-amperes. Express per unit impedance in terms of base MVA and base kV for three-phase systems. (CO1)
- (b) The one-line diagram for three generator systems is shown in Fig. Draw reactance diagram. Assume base values of voltage and MVA for low voltage side of 2 MVA transformer are 3 MVA and 11 kV. (CO1)



P. T. O.

- (c) A 25 kVA, 220/110 V three-phase transformer has a 0.04Ω leakage reactance referred to the low voltage side. Calculate the per unit leakage impedance referred to the low-voltage and high-voltage side of the transformer. Use the transformer rating as the base value. (CO1)

2. (a) Draw zero sequence network for the following transformers : (CO2)



- (b) Prove the following : (CO2)

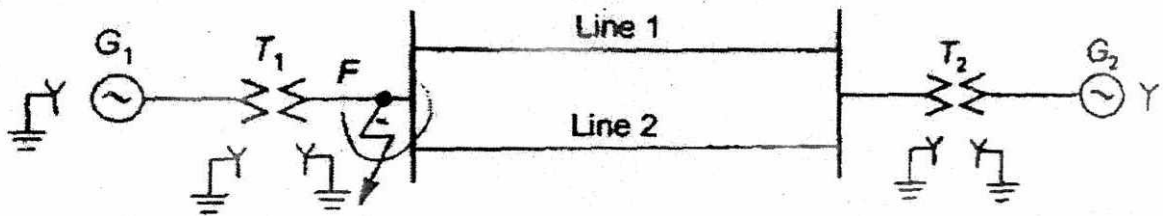
(i) The line voltage phasors V_{ab} , V_{bc} and V_{ca} will have no zero-sequence component.

(ii) Neutral current can flow only if zero-sequence currents exist.

- (c) Derive the expression of sequence impedance of a transmission line. Assume there is no any neutral impedance, self and mutual impedance of the line are Z_s and Z_m , respectively. (CO2)

3. (a) Discuss the various types of shunt faults existing in transmission systems. Also, draw the interconnection of sequence network in LL faults conditions. (CO3)

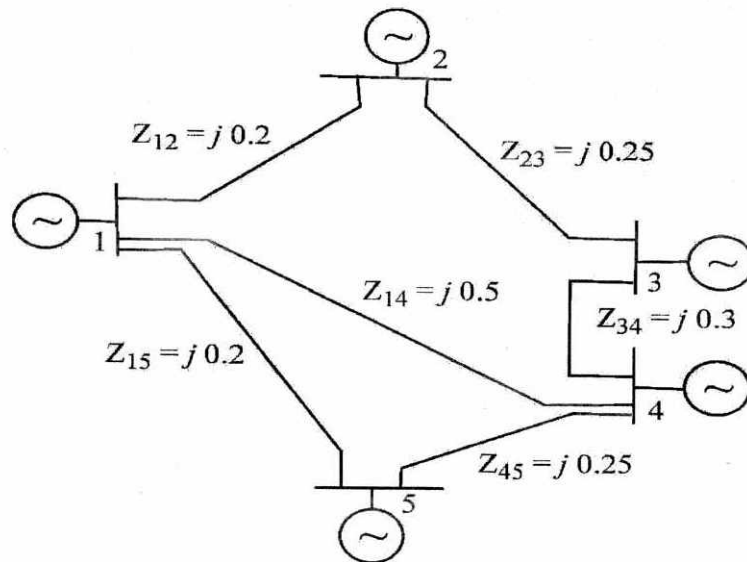
- (b) Determine positive, negative and zero sequence impedance for the power system as shown in figure below. The specification of the power system components (in p.u.) are indicated in table below : (CO3)



Components	X_0	X_1	X_2
Generator G_1	0.05	0.2	0.2
Generator G_2	0.03	0.25	0.15
Line 1	0.70	0.3	0.3
Line 2	0.70	0.3	0.3
Transformer T_1	0.12	0.12	0.12
Transformer T_2	0.10	0.10	0.10

- (c) A synchronous generator rated 500 kVA, 440 V, 0.1 p.u. subtransient reactance is supplying a passive load of 400 kW at 0.8 lag. p.f. Calculate the initial symmetrical r.m.s. current for a 3-phase fault at generator terminals. (CO3)
4. (a) Discuss the Gauss-Seidel method of load flow analysis. (CO4)

- (b) Determine Y_{bus} for the 5-bus system as shown in figure below. The line series impedances between the busses are indicated in figure : (CO4)



- (c) Derive static load flow equations and discuss the desirable feature of an ideal load flow analysis. (CO4)
5. (a) Define the following in terms with reference to the travelling wave : (CO5, CO6)
- Reflection coefficient
 - Refraction coefficient.
- Also derive the relation between reflection and refraction coefficient.
- (b) Consider a synchronous generator supplies a power to the infinite bus through double circuit line. Derive the expression of critical clearing angle, using equal area criterion, when faults are existing in one of the transmission line. (CO5, CO6)
- (c) A loss free generator supplies 50 MW to an infinite bus, the steady state limit of the system being 100 MW. Determine whether the generator will remain in synchronism if the prime mover is abruptly increased by 30 MW. (CO5, CO6)

Roll No.

TEE-603

B. TECH. (EE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

POWER SYSTEM PROTECTION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Define switchgear. Explain briefly the role of protection in a power system. (CO1)
- (b) Analyze the behaviour of a transmission line under sudden occurrence of fault and discuss doubling effect. (CO1)
- (c) On the occurrence of a fault, how is it cleared by the protective devices ? Draw a suitable diagram to support the answer. (CO1)
2. (a) Discuss the difficulties associated with differential protection. (CO2)

P. T. O.

- (b) Determine the time of operation of a 1A, 3s over current relay having plug setting of 125 percent and a time multiplier of 0.6. The supplying CT is rated 400 : 1 A and fault current is 4,000 A. The relay characteristic curve is given below : (CO2)

PSM	Time sec
1.3	30
2	10
4	5
8	3.3
10	3
20	2.2

- (c) Discuss the operating principle of Induction type relays. (CO2)
3. (a) Compare a static relay with electromagnetic relays. (CO3)
- (b) Explain Merz Price system of protection as applied to alternator. (CO3)
- (c) Discuss the main components of static relay. (CO3)
4. (a) Explain the terms (i) restriking voltage, (ii) recovery voltage, (iii) RRRV. (CO4, CO5)
- (b) In a short-circuit test on a 3-pole, 110 kV circuit breaker pf of the fault was 0.4, the recovery voltage was 0.95 times full-line value. The breaking current was symmetrical. The frequency of oscillation of restriking voltage was 15,000 c/s. Estimate the average rate of rise of restriking voltage. The neutral is grounded and the fault involves earth. (CO4, CO5)

(3)

- (c) Discuss Resistance Switching. In a 220 kV system, the reactance and capacitance up to the location of circuit breaker are 8 Ω and 0.025 μF respectively. A resistance of 600 ohm is connected across the contacts of the circuit breaker. Determine the following : (CO4, CO5)
- (i) Natural frequency of oscillation
 - (ii) Damped frequency of oscillation.
5. (a) What are the different types of circuit breakers when the arc quenching medium is the criterion ? Mention the voltage range for which a particular type of circuit breaker is recommended. (CO5, CO6)
- (b) Explain making capacity and breaking capacity of a circuit breaker. A three-phase circuit breaker is rated at 2000 MVA, 33 kV. Find its making current. (CO5, CO6)
- (c) Write a short note on SF_6 circuit breaker. (CO5, CO6)

Roll No.

TEE-605

B. TECH. (EEE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

DIGITAL CONTROL SYSTEMS

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) With the help of suitable circuit, explain the principle of operation of sample and hold circuit. Also, derive the transfer function of zero order hold circuit. (CO1, CO2)
(b) Consider a discrete time system with transfer function :

$$\frac{Y(z)}{R(z)} = \frac{1}{z^2 - \frac{3}{4}z + \frac{1}{8}}$$

Obtain the response $[y(k)]$ of the given transfer function for the following inputs $[r(k)]$: (CO1, CO2)

- (i) unit impulse input
(ii) unit step input.

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- (c) (i) Determine the z-transform of the sequence $x(n) = \{1 \ 2 \ -1 \ 0 \ 1 \ 0 \ 1\}$.
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- (ii) Determine the inverse z-transform of the following transfer function using partial fraction : (CO1, CO2)

$$X(z) = \frac{2z^3 + z}{(z-2)^2(z-1)}$$

2. (a) Define the following terms as associated with the state space representation : (CO3)
- (i) State
 - (ii) State Variable
 - (iii) State Vector
 - (iv) State Space
- (b) Obtain the transfer function of the system represented by the following state space : (CO3)

$$x(k+1) = Ax(k) + bu(k)$$

$$y(k) = cx(k) + du(k)$$

Also, give description of each element involved in the above representation.

- (c) Determine the state transition matrix (STM) of the following system : (CO3)

$$x(k+1) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

3. (a) What do you understand by controllability and observability ? Also, explain in brief, how the systems are tested for controllability and observability. (CO4)
- (b) Test the controllability and observability of the following system : (CO4)

$$x(k+1) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 4.4537 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -0.5809 & 0 & 0 & 0 \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ -0.3947 \\ 0 \\ 0.9211 \end{bmatrix} u(k)$$

$$y(k) = [0 \quad 0 \quad 1 \quad 0] x(k)$$

- (c) Investigate the controllability and observability of the following system : (CO4)

$$(i) \quad x(k+1) = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} x(k) + \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} x(k)$$

$$(ii) \quad x(k+1) = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} x(k) + \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} u(k)$$

$$y(k) = [1 \quad 1] x(k)$$

4. (a) Discuss the necessary and sufficient conditions for arbitrary pole placement. Also explain how the stability is improved through state feedback. (CO5)

- (b) What do you understand by state observer? Describe the steps to design a state observer for a linear time invariant system described by the following state space model : (CO5)

$$x(k+1) = Ax(k) + bu(k)$$

$$y(k) = cx(k) + du(k)$$

- (c) For the following transfer function : (CO5)

$$\frac{U(z)}{U(z)} = \frac{z+1}{z^2 + 1.3z + 0.4}$$

obtain :

- (i) Controllable canonical form
- (ii) Observable canonical form.

5. (a) Consider the problem of digital control of a plant described by the transfer function :

$$G(s) = \frac{1}{s+3}$$

The plant has a constant reference command signal. Design a PI control algorithm that results in system response characteristics : (CO6)

$$\zeta = 0.5 \text{ and } \omega_n = 5$$

- (b) Consider the problem of altitude control of a rigid satellite. The state equation of the plant is : (CO6)

$$x(k+1) = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} u(k)$$

Assume $T = 0.1$ sec and desired characteristics roots of the closed-loop system are $z_{1,2} = 0.875 \angle \pm 17.9^\circ$. Design a state regular if zeta and natural frequency of the system is 0.5 and 3.6 rad/sec respectively.

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TEE-605

- (c) Consider n th order single input single output system as expressed by the following state space model :

$$x(n+1) = Ax(n) + bu(n)$$

$$y(n) = cx(n)$$

Assume that the feedback is being used in the form :

$$u(n) = -kx(n) + r$$

where, ' r ' is the reference input signal.

Show that the zeros of the system are invariant under state-feedback.

TEE-605

Roll No.

TEE-608

B. TECH. (EE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023
HIGH VOLTAGE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Townsend's first and second ionization coefficients ? How is the condition for breakdown obtained in a Townsend discharge ?

(CO3, CO5)

- (b) State and explain conduction and breakdown phenomenon in commercial liquid.

(CO3, CO5)

- (c) Discuss the electromechanical breakdown of solid and obtain the equation for thermal breakdown.

(CO3, CO5)

2. (a) Discuss the difference between photo ionization and photo electric emission with suitable diagram.

(CO2, CO6)

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- (b) A solid specimen of dielectric has a dielectric constant of 4.4 and $\tan \delta = 0.001$ at a frequency of 50 Hz. If it is subjected to an electric field of 50 kV/cm. Calculate the heat generated in the specimen due to dielectric loss.
(CO2, CO6)
- (c) What do you understand by intrinsic strength of solid dielectric ? How does breakdown occur due to electrons in solid dielectric. (CO2, CO6)
3. (a) What is meant by control of transient or impulse voltages ? Why is this necessary ? (CO1, CO2)
- (b) Discuss Van de Graff Generator for high voltage dc generator with neat diagram. (CO1, CO2)
- (c) Explain the working of half wave and full wave rectifier for High voltage dc generation. (CO1, CO2)
4. (a) Draw the circuit for voltage multiplier circuit and explain its working with limitations. (CO1, CO6)
- (b) Explain the phenomenon of electrical conduction in liquids. How does it differ from that in gases ? (CO1, CO6)
- (c) Discuss the generation of high voltage ac using Tesla Coil. (CO1, CO6)
5. (a) Discuss multistage Impulse generation using Marx Circuit. (CO4, CO5)
- (b) Discuss measurement of high dc voltage using high ohmic series resistance. What are its limitations ? (CO4, CO5)
- (c) Discuss the techniques applied to measures the ripple in high voltage dc system. (CO4, CO5)